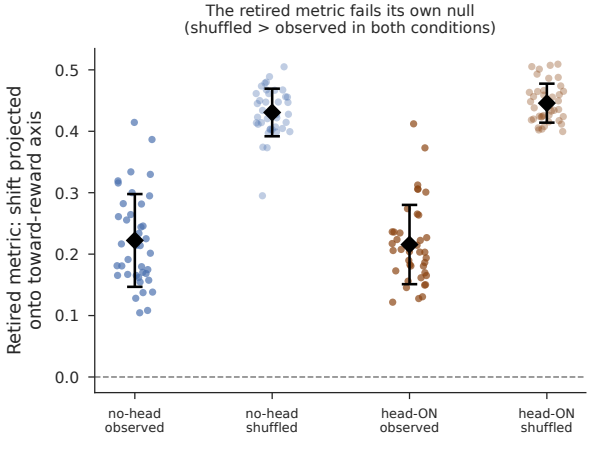
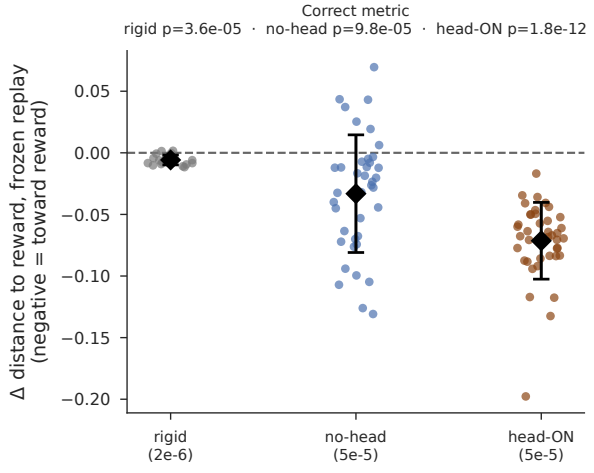
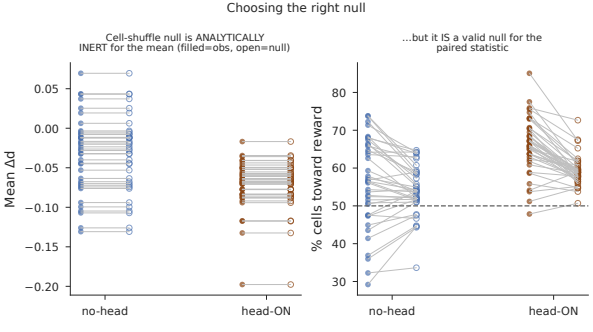
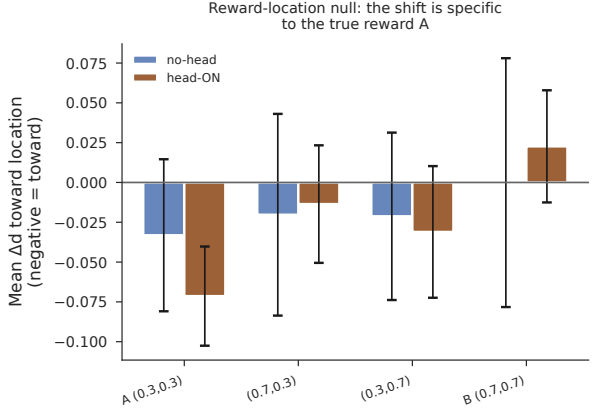

The retired metric fails its own null
(shuffled > observed in both conditions)
Retired metric: shift projected onto toward-reward axis
0.5
0.4
0.3
0.2
0.1
0.0
no-head observed
no-head shuffled
head-ON observed
head-ON shuffled


Correct metric
rigid p=3.6e-05 · no-head p=9.8e-05 · head-ON p=1.8e-12
Δ distance to reward, frozen replay
(negative = toward reward)
0.05
0.00
−0.05
−0.10
−0.15
−0.20
rigid (2e-6)
no-head (5e-5)
head-ON (5e-5)


Choosing the right null
Cell-shuffle null is ANALYTICALLY INERT for the mean (filled=obs, open=null)
Mean Δd
0.05
0.00
−0.05
−0.10
−0.15
−0.20
no-head
head-ON
...but it IS a valid null for the paired statistic
% cells toward reward
80
70
60
50
40
30
no-head
head-ON


Reward-location null: the shift is specific to the true reward A
no-head
head-ON
Mean Δd toward location
(negative = toward)
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
A (0.3,0.3)
(0.7,0.3)
(0.3,0.7)
B (0.7,0.7)